DialectGuard: Evaluating Response-Risk-Triggered Retrieval for Factual Question Answering Across Arabic Varieties

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Abstract

Factual question answering may vary across Arabic linguistic varieties, motivating interventions that combine dialect normalization and retrieved evidence. This study evaluated DialectGuard, a heuristic framework that uses disagreement among three language models and a non-answer signal to trigger normalization, retrieval, and grounded regeneration. A review-locked held-out candidate benchmark comprised 100 factual questions expressed in Modern Standard Arabic, Jordanian, Egyptian, and Saudi Arabic. Four conditions produced 1,600 held-out responses; 2,420 additional experimental records supported development, controlled comparisons, ablations, and efficiency assessment. DialectGuard flagged 378 of 400 held-out variants (94.5%) and had a median recorded latency of 16.49 seconds, compared with 7.70 seconds for always-on retrieval. In 318 paired cases with determinate AI-assisted correctness annotations, the observed difference from always-on retrieval was zero percentage points (question-cluster bootstrap 95% interval: −5.8 to 6.0). However, 278 annotations remained ambiguous after provisional resolution, and claim-level evidence verification was incomplete. The tested architecture therefore did not demonstrate a latency advantage or verified factual superiority. The findings emphasize the importance of inexpensive risk detection, controlled generation comparisons, and independent Arabic-language adjudication.

Keywords: Arabic dialects; factual question answering; retrieval-augmented generation; risk detection; hallucination evaluation; large language models

Introduction

Large language models are increasingly used to answer factual questions, yet fluent responses can be incomplete, misleading, or incorrect. Retrieval-augmented generation (RAG) introduces external documents into the generation process, but retrieval itself introduces computational overhead and does not establish that selected passages are true. The reliability–cost trade-off becomes particularly relevant when users pose straightforward questions that may not require external evidence. Furthermore, the additional task of deciding when retrieval is useful can itself be expensive if a risk detector relies on multiple large models.

Arabic provides an informative setting for this investigation. Factual intent may be expressed in Modern Standard Arabic (MSA) or in regionally marked forms that differ lexically and syntactically. A question that names the same historical event, location, or scientific entity may elicit different model answers after a change in linguistic variety. DialectalArabicMMLU documented differences in Arabic-model performance across dialects using a comparatively large manually adapted benchmark (Altakrori et al., 2026). Such evidence motivates targeted analysis of factual answer reliability, while cautioning against generalizations from any one small constructed set.

The evaluation of factuality also requires care beyond string matching. A correct response may use an acceptable Arabic transliteration or synonym instead of the exact reference phrase. A response may also supply the required entity while adding an erroneous date or unsupported detail. Alansari and Luqman (2025) proposed a fine-grained Arabic hallucination evaluation framework, underscoring the distinction between different factual failure types. Accordingly, the present study distinguished preliminary answer correctness from independently verified claim-level factual support, rather than collapsing the two into a single automatically calculated measure.

Prior adaptive retrieval methods typically decide whether and how to retrieve according to task demands. Adaptive-RAG chose strategies according to question complexity (Jeong et al., 2024); Self-RAG incorporated retrieval-related reflection into generation (Asai et al., 2024); and FLARE considered retrieval during generation based on anticipated content (Jiang et al., 2023). The present system examined a different operational trigger: disagreement between model responses together with baseline non-answer behavior. This comparison is a difference in mechanism, not a claim that the present method is novel in all respects or superior to prior adaptive systems.

The central hypothesis was that a response-risk signal could identify enough problematic baseline answers to

make selective mitigation attractive, while leaving lower-risk responses untouched. Five research questions guided the study: (RQ1) How do factual-answer outcomes vary across four Arabic varieties? (RQ2) Does a disagreement-based detector identify baseline answer failures? (RQ3) Does MSA normalization change correctness? (RQ4) How do retrieval and adaptive regeneration compare against unretrieved generation? (RQ5) Does adaptive intervention reduce observed computational costs relative to always-on RAG? The following sections describe a completed experimental implementation and distinguish what its results demonstrate from claims that remain unverified due to annotation uncertainty.

Materials And Methods

Research design and question sets

The experiment used an Arabic factual question-answering benchmark constructed from source-based questions and adapted into four varieties: MSA, Jordanian, Egyptian, and Saudi Arabic. The development partition contained 65 base questions with four varieties each (260 variants). A separate held-out candidate partition contained 100 base questions with four varieties each (400 variants). All four experimental conditions were run for each held-out variant. Four varieties derived from a single base question were treated as a dependent cluster in uncertainty estimation, not as four independent factual tasks.

The held-out partition was designated review-locked v0.1. The available preparation records included review and filtering steps, but complete independent dialect-speaker signoff and full semantic-overlap certification were not documented. The private gold reference file was separated from model inference. The benchmark should thus be understood as an internally prepared experimental test set rather than an externally validated population-representative Arabic QA dataset.

Models and retrieval corpus

The baseline generator was Qwen3-4B-4bit; Gemma-3-4B-it-4bit and Granite-4.0-H-3B-ar supplied comparison responses for risk assessment. Granite was also used for grounded responses in the primary retrieved conditions. Multilingual MiniLM representations supported dense evidence retrieval, followed by cross-encoder reranking. The frozen development-derived evidence corpus contained 37 relevance-screened Arabic Wikipedia passages from 31 URLs. Relevance screening was not equivalent to independent verification of factual correctness; coverage of held-out queries was not guaranteed.

Interventions and routing

Four primary conditions were compared. Baseline used Qwen on the original input. Normalization transformed the question into MSA before Qwen generation. Always-RAG retrieved and reranked passages before generating an answer with Granite. DialectGuard first collected responses for disagreement detection; higher-risk questions were normalized, passed through retrieval and reranking, and answered using Granite, whereas lower-risk questions retained the Qwen response. The heuristic increased the risk score when answer similarity was below its threshold and when the baseline produced a non-answer; a fixed risk threshold triggered the intervention. This score was neither a probabilistic hallucination estimate nor a calibrated confidence score.

The principal comparison between baseline and the retrieved conditions involved both retrieval and a change in generator (Qwen versus Granite). Consequently, the experiment did not fully identify a causal effect of retrieval alone. A supplementary controlled-Qwen study, along with component ablations and a separate timing benchmark, was conducted to probe implementation differences. Their existence does not remove the original confounding or validate generated facts.

Experimental execution and quality control

The code was developed for an Apple Silicon Mac M2 with 16 GB unified memory, using sequential model loading to reduce memory pressure. The experiment repository preserved model revision information, a frozen configuration, output manifests, and result-level logs. A total of 1,040 development, 520 controlled-Qwen, 780 ablation, 80 efficiency, and 1,600 held-out rows were reported. The held-out results contained 400 rows per condition, no duplicated question–condition pairs, and zero reported execution errors. Eight final generated answers were empty despite successful run status: five in always-RAG and three in DialectGuard. These were retained as non-correct abstentions for accuracy reporting.

Answer assessment, missingness, and statistical analysis

Held-out responses were assessed through AI-assisted screening, rule-based revisions, and researcher review of an earlier annotation version. Responses were categorized as correct, incorrect, abstention, empty, or ambiguous. A later question-specific alias/entity resolution pass proposed labels for 346 initially

ambiguous responses, leaving 278 unresolved. The later changes did not have documented independent expert adjudication. Thus, the label-based analyses below are exploratory. Correctness was distinguished from claim-level evidence support; evidence-support and hallucination rates were not declared verified without claim-by-claim source evaluation.

For each condition, determinate accuracy was calculated as correct divided by the number of responses not labeled ambiguous, with abstentions and blanks counted as non-correct. All-sample identification bounds were computed by assigning all ambiguous cases first to incorrect and then to correct. Matched-condition effects were calculated only for variants with determinate labels in both conditions. Base-question-clustered bootstrap intervals (rather than question-variant independent intervals) accounted for repeated dialect expressions. Exploratory discordance tests were reported without multiple-comparison correction; neither these tests nor missing-at-random assumptions establish population causal effects.

Results And Discussions

Experiment integrity and intervention frequency

The repository yielded 4,020 recorded experimental outputs across five experiment groups. The held-out evaluation generated 1,600 rows from 400 question variants under four conditions. The intervention rule classified 378 variants as high risk and 22 as low risk, a routing rate of 94.5% (Wilson 95% interval approximately 91.8%–96.3%). This left limited opportunity for the approach to avoid retrieval and regeneration overhead. The risk flag should not be confused with an observed incorrect-answer rate.

Table 1. Distribution of completed experimental records

Experiment	Recorded rows
Development: four conditions	1,040
Controlled Qwen retrieval comparison	520
Component ablations	780
Efficiency benchmark	80
Held-out: four conditions	1,600
Total	4,020

Provisional response correctness and uncertainty

The provisionally resolved label dataset yielded 132 correct baseline responses, 122 correct normalized responses, and 145 correct responses in each of always-RAG and DialectGuard. The determinate subsets comprised 326, 326, 335, and 335 responses, respectively. The resulting condition estimates were therefore not directly comparable as four equally complete sets. Across the full 400-variant test set per condition, bounds remained wide because the 278 unresolved responses were not assigned a correctness value.

Table 2. Provisional held-out correctness under incomplete adjudication

Condition	Correct	Incorrect	Ambiguous
Baseline	132	194	74
Normalization	122	204	74
Always-RAG	145	138	65
DialectGuard	145	127	65

Table 2 (continued): Abstentions, accuracy and uncertainty bounds.

Condition	Abstain / empty	Determinate acc.	Full-set bound
Baseline	0	40.5%	33.0–51.5%
Normalization	0	37.4%	30.5–49.0%
Always-RAG	52	43.3%	36.3–52.5%
DialectGuard	63	43.3%	36.3–52.5%

Note. The results rely on AI-assisted classifications and unverified later rule-based revisions. Full-set bounds are identification ranges, not sampling confidence intervals. Blanks were included in abstain/empty.

Crucially, 43.3% determinate accuracy for DialectGuard and always-RAG did not constitute evidence of equivalent performance over all 400 variants; the two systems differed in which responses were determinate. Even the common-subset analysis was conditional on successful adjudication and could be affected by systematic ambiguity. An observed advantage based only on a selectively retained subset could therefore misrepresent the complete benchmark.

Paired condition comparisons

In the 318 held-out variants that had determinate labels for both DialectGuard and always-RAG, the measured mean paired correctness difference was exactly 0.0 percentage points. The base-question-cluster bootstrap 95% interval extended from −5.8 to +6.0 percentage points. A zero point estimate with this interval did not establish equivalence or non-inferiority. The corresponding DialectGuard-minus-baseline point estimate was +3.7 percentage points among 294 evaluable pairs, with an interval spanning both negative and positive differences. None of these exploratory comparisons justified a conclusion of superior factual accuracy.

Table 3. Paired complete-case correctness contrasts

Contrast	Paired n	Difference (pp)	Cluster 95% interval (pp)
DialectGuard − Baseline	294	+3.7	−7.6 to +14.8
DialectGuard − Always-RAG	318	0.0	−5.8 to +6.0
Always-RAG − Baseline	292	+2.4	−9.3 to +14.2
Normalization − Baseline	314	−1.9	−7.1 to +3.7

Note. All comparisons were restricted to pairs with determinate AI-assisted labels; uncertainty is exploratory and conditional on adjudication.

Detector behavior

Within the 326 baseline questions with determinate correctness labels, the detector produced 193 true-positive high-risk predictions, 117 false positives, one false negative, and 15 true negatives when "error" referred to an incorrect baseline answer. Conditional precision was 62.3%, sensitivity was 99.5%, and specificity was 11.4%. The pattern combined very high sensitivity with limited selectivity. Because 74 baseline responses remained ambiguous, these were conditional diagnostic values, not full-test detector characteristics. In particular, the detector's operational expense might outweigh the savings associated with the 22 low-risk cases.

Dialect-level patterns

Preliminary condition-by-variety summaries suggested that evaluated-answer correctness varied between MSA, Jordanian, Egyptian, and Saudi Arabic; however, the patterns were not uniform across conditions. For example, among the determinate DialectGuard answers the Egyptian, Jordanian, MSA, and Saudi varieties yielded 32/81, 39/85, 37/85, and 37/84 correct, respectively. The limited number of questions, shared base-question content, and nonuniform ambiguity rates prevent these differences from supporting robust conclusions about dialect populations or relative linguistic difficulty.

Table 4. DialectGuard correctness by Arabic variety

Arabic variety	Correct	Determinate	Provisional accuracy
Egyptian	32	81	39.5%
Jordanian	39	85	45.9%
MSA	37	85	43.5%
Saudi	37	84	44.0%

Note. Differences are descriptive and are influenced by unresolved annotation counts.

Latency and realized computational trade-off

Median recorded held-out latency was 1.954 seconds for baseline, 4.420 seconds for normalization, 7.705 seconds for always-RAG, and 16.491 seconds for DialectGuard. DialectGuard was slower than always-RAG in 399 of 400 paired runs. The median paired excess latency was 8.735 seconds. When observations were grouped by the 100 base questions, the median base-level average excess was 8.382 seconds, with an approximate cluster bootstrap 95% interval of 8.201–8.572 seconds. Thus, the hypothesized efficiency advantage was not realized in the tested implementation.

Table 5. Observed held-out median latency

Condition	Median seconds
Baseline	1.954
Normalization	4.420
Always-RAG	7.705
DialectGuard	16.491

Note. Hardware- and implementation-dependent recorded timings; model setup and retrieval-index construction were not fully included in each question's elapsed time.

This result is interpretable in light of the detector design. Risk assessment required multiple model responses and triggered high-cost mitigation for nearly all variants. Consequently, adaptive routing incurred the detector's expense while rarely avoiding the downstream RAG path. The study suggests that future selective retrieval systems must account for detector inference cost explicitly; low-cost risk predictors or earlier stopping rules may be essential to obtaining a favorable trade-off.

Interpretation and validity threats

The completed experiments demonstrate that risk-triggered Arabic QA can be implemented and logged reproducibly under a fixed local configuration. They do not demonstrate verified factual improvement over unconditional retrieval. The absence of claim-level evidence adjudication prevents a trustworthy hallucination-rate comparison. Even where a proposed annotation labels a response correct, its accompanying explanation may contain unsupported claims. Additionally, a model's inability to use a particular reference passage does not by itself mean that the final answer is false.

The remaining limitations are substantial: a relatively small and internally adapted question set; imperfect documentation of dialect review and semantic independence; an evidence corpus derived from development tasks; differences in answer generators between conditions; incomplete independent annotation; and latency tied to local model loading and hardware. These issues limit the study's causal, linguistic, and operational generalizability. The purpose of including these caveats was to keep the conclusions proportionate to the evidence rather than to characterize experimental completion as factual verification.

Conclusions & Recommendations

The completed DialectGuard study evaluated four conditions across 400 held-out Arabic question variants and documented 4,020 experimental records overall. The detector routed almost every held-out input through its mitigation path, and the measured

adaptive pipeline was slower than always-on retrieval. Preliminary, incomplete factual labels did not establish a reliable accuracy advantage for the adaptive condition. The findings therefore did not support the hypothesized combination of better factual reliability and lower computational cost in the examined configuration.

Future work should prioritize a low-cost, calibrated detector; a single common answer generator across all conditions; broader, independently validated Arabic dialect data; and claim-level evidence verification with documented annotator agreement. Future experiments should also report confidence intervals and coverage for uncertain labels rather than silently excluding difficult responses. The present paper provides a transparent experimental baseline and a practical account of why selective mitigation cannot be assumed to be computationally efficient.

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Appendices

Appendix A. Experimental conditions and model roles

Baseline: Qwen answered the input question without retrieval.

Normalization: the question was converted into Modern Standard Arabic before Qwen response generation.

Always-RAG: passages were retrieved and reranked before Granite generated the response.

DialectGuard: model-disagreement and Qwen non-answer signals determined whether normalization, retrieval, and Granite regeneration were triggered.

Appendix B. Dataset and evaluation scope

Development: 65 base questions in four varieties (260 variants); four main conditions yielded 1,040 records.

Held-out: 100 base questions in four varieties (400 variants); four conditions yielded 1,600 records.

Additional experiments: controlled Qwen RAG (520 records), ablations (780), and efficiency benchmarking (80).

Evaluation outcomes were categorized as correct, incorrect, abstention/empty, or ambiguous. Provisional rule-based adjudication left 278 responses ambiguous.

Appendix C. Reproducibility and provenance

Repository: https://github.com/MazenZeybek/DialectGuard

Frozen held-out inference commit: 12cd956c5a1afc11b3a156c1e64f8d863e97a09a.

Recorded files include heldout_experiment_results.csv, heldout_run_manifest.json, heldout_quality_control.json, the development, controlled-comparison, ablation, and efficiency CSVs.

Held-out risk decisions: 378 high-risk and 22 low-risk variants; eight empty final responses must remain in the evaluation denominator.

The development retrieval evidence corpus contained 37 relevance-screened passages from 31 Arabic Wikipedia URLs, without independent claim-level factual certification.

Appendix D. Annotation and inference limitations

Factual labels incorporate AI-assisted screening and rule-based reassignment. The final 346 rule-based reclassifications were not independently certified.

Difference estimates based on determinate pairs are conditional on missing adjudication labels; these are not full-benchmark superiority estimates.

The principal baseline and retrieved conditions differ in answer generator as well as retrieval; controlled Qwen experiments partly investigate that confounding.

Per-question latency excludes initial model loading and index creation; observed timings are local-hardware dependent.